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In [1]: from sklearn.metrics.pairwise import cosine_similarity
import pandas as pd
import numpy as np
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.metrics.pairwise import cosine_similarity
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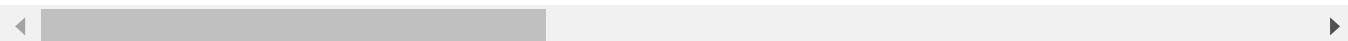
```
In [3]: df = pd.read_csv(r"D:\College\TE\SEM-2\Practical\DSBDA\MP1\movie_dataset.csv")
```

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In [4]: df.head()
```

Out[4]:

	index	budget	genres	homepage	id	keywords	original_language
0	0	237000000	Action Adventure Fantasy Science Fiction	http://www.avatarmovie.com/	19995	culture clash future space war space colony so...	
1	1	300000000	Adventure Fantasy Action	http://disney.go.com/disneypictures/pirates/	285	ocean drug abuse exotic island east india trad...	
2	2	245000000	Action Adventure Crime	http://www.sonypictures.com/movies/spectre/	206647	spy based on novel secret agent sequel mi6	
3	3	250000000	Action Crime Drama Thriller	http://www.thedarkknighttrises.com/	49026	dc comics crime fighter terrorist secret ident...	
4	4	260000000	Action Adventure Science Fiction	http://movies.disney.com/john-carter	49529	based on novel mars medallion space travel pri...	

5 rows × 24 columns



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In [5]: features = ['keywords', 'cast', 'genres', 'director']
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In [6]: def combine_features(row):
return row['keywords']+" "+row['cast']+" "+row['genres']+" "+row['director']
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In [7]: for feature in features:
df[feature] = df[feature].fillna('')

df["combined_features"] = df.apply(combine_features,axis=1)
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In [8]: cv = CountVectorizer()
count_matrix = cv.fit_transform(df["combined_features"])
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In [9]: cosine_sim = cosine_similarity(count_matrix)
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In [10]: def get_title_from_index(index):
          return df[df.index == index]["title"].values[0]
def get_index_from_title(title):
    return df[df.title == title]["index"].values[0]
```

```
In [11]: movie_user_likes = "Avatar"
movie_index = get_index_from_title(movie_user_likes)
similar_movies = list(enumerate(cosine_sim[movie_index]))
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In [12]: sorted_similar_movies = sorted(similar_movies, key=lambda x:x[1], reverse=True)[1:]
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In [13]: i=0
print("Top 5 similar movies to "+movie_user_likes+" are:\n")
for element in sorted_similar_movies:
    print(get_title_from_index(element[0]))
    i=i+1
    if i>5:
        break
```

Top 5 similar movies to Avatar are:

Guardians of the Galaxy
Aliens
Star Wars: Clone Wars: Volume 1
Star Trek Into Darkness
Star Trek Beyond
Alien